MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 1	Unit 3 - 3.1 Web Interface for the Game (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Unit 2 test avg 76%; selection tools weakest. Short re-teach sheet sent. New unit: start from students' game ideas from homework.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students set up the 'Bit by Bit' worm-game project: folder structure, naming (camel/kebab case), VS Code, and the HTML structure of the index page. Sprint 3.1 obj. 1; ISTE 4.a; Mass. 9-12.DTC.a.1; CSTA 3A-AP-16.

Success criteria — *what will successful learning look / sound like?*

I create the correct folder structure (html, css, js, images, audio); I name files in kebab case; I write a valid index.html with head, body, header, sections and form elements.

Key knowledge / skill / understanding:

HTML structures content; folders keep projects organised; naming conventions avoid broken links.

Likely misconception / access or challenge need:

Spaces in file names are fine; HTML styles the page; EAL: 'hierarchy', 'kebab case'.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Brainstorming **Time:** 7 min

What students will do:

Groups brainstorm what a 'catch the worm' vermicompost game needs on the start page (title, player name, glove choice, start button) and sketch the layout.

Evidence tool: Observation checklist

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

A3 paper/markers; vermicompost picture (real-life link).

Support (low performers / SEN):

Picture prompts of game screens.

Stretch (fast learners / gifted):

Add an accessibility feature to the sketch.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Download the resource folder; create folders and kebab-case files; in VS Code write index.html structure (doctype, head, title, header, main, form, input, button) following the teacher.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; resource folder go.qubitscs.com/cXAd; QR LCdB; source file h9pt.

Support (low performers / SEN):

Starter index.html with comments marking where to type.

Stretch (fast learners / gifted):

Add semantic tags and alt text; validate the page with the W3C validator.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 8 min

What students will do:

Pairs examine 3 file names and 3 code snippets; decide which follow camel/kebab case and which HTML has errors, then share fixes.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with snippets; mini-whiteboards.

Support (low performers / SEN):

Highlighted lines to focus on.

Stretch (fast learners / gifted):

Rewrite a snippet using semantic HTML.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: play the finished worm game for 1 minute. Real-life link: vermicompost (earthworms recycle food waste - school garden). Model folder setup and VS Code basics (Explorer, Live Server/open in browser). Key words: camel case, kebab case, hierarchy, element, tag.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students build a real browser game across Units 3-4, linked to sustainability (vermicompost) - a project that grows each lesson.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Files saved in Downloads not the project folder; missing closing tags; uppercase/space file names; students copying the source file without understanding. Response: 'my page is blank' - file not saved/opened.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write the correct folder tree and one kebab-case file name; one question for the teacher. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: complete Challenge 1 page-structure sketch for a personal website (drawing + 5 HTML tags you would use).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Game demo hooked everyone; vermicompost link sparked talk about recycling practices in different home countries. Folder structure done by 90%.

What will I change next time?

Check VS Code is installed on all lab PCs before the unit (lost 5 minutes in 10/4). Keep the starter file for SEN.

Students needing follow-up:

Students new to coding (mainly recent arrivals) need a 'tags cheat sheet'; 3 students did not save in the right folder.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☑1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 2	Unit 3 - 3.1 Web Interface for the Game (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 88% correct folder tree; closing-tag errors common. Today: start with a 'spot the bug' and build the game page.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students build the HTML structure of the game page (score, worm count, holes grid, tray, gloves, buttons) and plan a page structure from a layout. Sprint 3.1 obj. 1; Challenges 2-3; CSTA 3A-AP-17.

Success criteria — *what will successful learning look / sound like?*

I break a layout into header/container divs; I write the game page HTML with ids/classes for each part; my page opens without errors.

Key knowledge / skill / understanding:

div containers, id vs class, nesting creates hierarchy; plan structure before coding.

Likely misconception / access or challenge need:

id and class are the same; every element needs its own CSS file.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 8 min

What students will do:

Challenge 2: convert a given web page screenshot into a page-structure diagram (header, innerHeader1/2, container, innerContainer1/2).

Evidence tool: Open-ended task

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Printed screenshot; diagram template.

Support (low performers / SEN):

Template with boxes pre-drawn.

Stretch (fast learners / gifted):

Add class names using camel case.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code game.html: header with score and worm count, a holes container with 9 holes (div), worm tray, glove area, and buttons, each with ids/classes as in the source file.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; QR nCfN; source file F9aL.

Support (low performers / SEN):

Partially completed game.html; colour-coded nesting guide.

Stretch (fast learners / gifted):

Use a loop-friendly naming pattern (hole1-hole9) and comment each section.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Jigsaw **Time:** 7 min

What students will do:

Expert groups each explain one part of the page (header, holes, tray, gloves) to a mixed group, using their code.

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Each student's code; jigsaw cards.

Support (low performers / SEN):

Expert role card with prompts.

Stretch (fast learners / gifted):

Explain how this structure will help JavaScript later.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Spot the bug: 3 HTML lines with missing closing tags. Model planning on paper first: draw boxes for the game page, then translate into nested divs. Explain id (unique) vs class (shared). Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Jigsaw: every student becomes an 'expert' explaining one part of the game structure - builds learner voice and collaboration across language groups.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Mis-nested divs; duplicate ids; students not linking files later; copying without understanding - ask 'what will this div hold?'. Likely response: 'Can I just use class for everything?'

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: label the structure diagram of the game page with correct tags; learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: Challenge 3 - draw a responsive image-gallery layout and write its HTML (no CSS yet).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Planning on paper first reduced nesting errors a lot. Jigsaw groups mixing Arabic, Urdu and English speakers worked; experts were proud to explain.

What will I change next time?

Give more time for the game page; 30% needed the starter file. Shorten jigsaw to 2 rounds.

Students needing follow-up:

10/2 and 10/6: 4 students with duplicate ids to fix. Gifted coders could start the gallery challenge in class.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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